Exercise Sheet 9

Anomaly Detection

Hint: Commit your code to your repository regularly as you work through the exercises - a clean, meaningful commit history is part of your final submission.

Out-of-Distribution Detection

Exercise 9.1: The OOD Problem

Your CARLA model was trained on sunny daytime images. At deployment, it may encounter images from fog or night - conditions outside its ODD.

1. Why can a standard classifier not be trusted to signal when it receives an OOD input?

Standard classifiers are built under a **closed-world assumption**, meaning they are trained to distribute probability mass exclusively among a fixed set of known classes.

- **The Softmax Bottleneck:** The final softmax layer normalizes raw logit outputs so that they sum to 1. Even if an input is completely anomalous or outside the operational design domain (such as unexpected fog or night conditions), the network is forced to pick from its predefined categories.
- **Mathematical Amplification:** If one random class logit happens to be slightly larger than the others—even if all activations are incredibly weak—the exponential nature of the softmax function will amplify it into a deceptively high-confidence prediction.
- **Lack of Calibration:** Standard models are optimized to minimize empirical risk, not to estimate true epistemic uncertainty. They fundamentally do not "know what they don't know."

2. Why is silent failure (confidently wrong prediction) worse than uncertain failure in a safety-critical system?

In safety-critical applications like autonomous driving, the main difference between a controlled situation and a catastrophe is whether the system can trigger its fail-safes.

- **Uncertain Failure (Manageable):** If a model encounters a strange input and flags it with low confidence, downstream safety monitors can detect this high uncertainty. The system can then initiate a safe fallback strategy—such as slowing down, pulling over, or handing control back to a human operator.
- **Silent Failure (Confidently Wrong):** When a model fails silently, it outputs an incorrect prediction with dangerously high confidence. Because downstream path planning and control modules receive what looks like a "certain" signal, no safety alerts are raised. For instance, the vehicle might confidently perceive a clear road when an obstacle is actually hidden in the fog, causing it to continue at full speed without braking.

Exercise 9.2: Baseline OOD Detection

Describe the maximum softmax probability (MSP) baseline for OOD detection. How does it work, and what are its main limitations?

Description of the Maximum Softmax Probability (MSP) Baseline

The Maximum Softmax Probability (MSP) baseline is the standard starting point for out-of-distribution detection.

- **How it works:** It operates on the assumption that a network will naturally be more confident when classifying data that looks like its training set (In-Distribution) than data that looks entirely alien (Out-of-Distribution). For every input image, the model calculates standard class probabilities, and the highest single probability value (the maximum softmax score) is used as the OOD score. If this score falls below a predetermined threshold, the input is flagged as OOD.

Main Limitations of MSP

While convenient because it requires no architectural modifications or model retraining, MSP suffers from major drawbacks:

1. **Severe Overconfidence:** Deep neural networks naturally map points that are far away from the training distribution to high-magnitude logits, leading to dangerously high confidence scores even on completely unrepresentative or garbage data.

2. **Loss of Absolute Scale:** Softmax only looks at the *relative* differences between logits, completely discarding their absolute magnitudes. An input could cause incredibly weak activations across the board (meaning the network doesn't genuinely recognize anything), but if one logit is slightly larger than the rest, MSP will still output a high-confidence score.

3. **Discriminative Alignment:** The model's internal features are optimized purely to tell the difference *between* in-distribution classes, making them inherently suboptimal for identifying when an input doesn't belong to *any* of those classes.

Exercise 9.3: Alternative Methods

Describe one more advanced OOD detection method (e.g., energy score or Mahalanobis distance). How does it improve over MSP?

To overcome the fundamental limitations of the Maximum Softmax Probability (MSP) baseline, advanced OOD detection techniques avoid relying on the highly compressed, relative probabilities of the final softmax layer. A widely used and powerful alternative is **Mahalanobis Distance-Based OOD Detection**.

How It Works

Instead of looking at the network's final class predictions, this method examines the **deep feature representations** (the embeddings) extracted from the core hidden layers of the neural network.

1 **Modeling In-Distribution Data:** It assumes that the intermediate features $f(x)$ of the in-distribution training data follow a multivariate Gaussian distribution for each class.

2 **Computing Statistics:** Using the training dataset, it calculates the empirical mean vector μ_c for each class c, along with a shared (tied) covariance matrix Σ across all classes to define the shape of the in-distribution feature space.

3 **Scoring Test Inputs:** When a new test image arrives, the network extracts its feature vector $f(x)$ and computes its Mahalanobis distance to the closest class mean:

$$d_M(x) = \min_c \sqrt{(f(x) - \mu_c)^T \Sigma^{-1} (f(x) - \mu_c)}$$

A higher distance score indicates that the input does not structurally match any known in-distribution cluster, flagging it as OOD.

How It Improves Over MSP PDF

- **Bypasses the Softmax Bottleneck:** MSP relies on relative logit differences scaled to sum to 1, completely throwing away absolute magnitude. Mahalanobis distance measures the *absolute* position of an input in feature space. If an anomalous input (like a dark, foggy night scene) yields completely unrecognizable features, it will land far away from all valid training clusters and be cleanly flagged—regardless of what the final softmax layer tries to guess.
- **Accounts for Covariance and Feature Variance:** By incorporating the covariance matrix Σ, the detector scales distances based on how the training data actually varies. It recognizes that some directions in feature space naturally have higher variance, avoiding false positives while remaining highly sensitive to anomalous deviations.
- **Leverages Higher-Dimensional Context:** The hidden layers of a network contain rich, high-dimensional geometric context about textures, shapes, and semantic layouts. MSP compresses all of this nuanced information down into a simple 1D vector of class scores, whereas feature-based distance tracking preserves this underlying domain data to catch subtle distribution shifts.

Practical: OOD Detection for the CARLA Model

The provided dataset includes images collected under conditions outside your training distribution: *fog* and *night*. It also includes images captured in a *different CARLA town* (different building layout, road geometry, and vegetation) under otherwise nominal sunny-daytime conditions. The fog and night images serve as clear OOD test data; the different-town images are examined separately below.

Exercise 9.4: Visualising the Distribution Shift

1. Display five in-distribution (sunny / daytime) images next to five fog and five-night images.
2. Also display five images from the different CARLA town. How do they differ from your training images, and how does that difference compare to the fog / night shift?
3. Compute the mean softmax confidence of each of your three models on in-distribution vs. fog/night images. Is the model more or less confident on these inputs?

Exercise 9.5: Is the Different Town Out-of-Distribution?

The different-town images were taken under nominal weather and lighting - the kind of conditions your system is meant to handle - but in a location your models never saw during training.

1. Look back at the ODD you specified in Exercise 2.2. As written, does your ODD clearly decide whether the different-town images are inside or outside it?

Ambiguity of the Original ODD Definition

The original Operational Design Domain (ODD) specified in Exercise 2.2 does not clearly decide whether the different-town images are inside or outside the domain boundaries.

- Under the **Scene Type** dimension, the valid operating condition is broadly defined as "Urban roads, mapped intersections".
- The different-town images consist of urban roads, distinct road geometries, and varied vegetation captured under nominal sunny-daytime conditions.
- Because these images represent nominal weather conditions on urban roads, they superficially fulfill the original "valid" condition text.
- However, these images originate from a completely separate CARLA town that the perception models never encountered during training.
- The original ODD fails to clarify whether "mapped" implies any globally available simulator map or if it is strictly restricted to the exact geographic environments present in the training distribution. This omission leaves the spatial boundaries fundamentally ambiguous.

2. Revise the wording of your ODD so that the new town is *unambiguously* either inside or outside it. State which you chose and why.

Dimension	Operating Condition (Valid)	Non-Operating Condition	Runtime Detection Method
Weather	Clear, cloudy (no rain/fog).	Heavy rain, fog, snow.	Anomaly/OOD detection scores.
Lighting	Daytime (high sun angle).	Night, low sun (glare).	Average pixel intensity.
Camera	Forward-facing, rigidly fixed.	Loose mount, lens obscuration.	Edge detection/Explainability.
Scene Type	Urban roads, mapped intersections.	Off-road, unmapped highways.	Scene classifier/OOD scores.
Vehicle Speed	<= 50-km/h.	> 50-km/h.	Vehicle speedometer feedback.

Revised ODD Dimension: Scene Type

Dimension	Operating Condition (Valid)	Non-Operating Condition	Runtime Detection Method
Scene Type	Urban roads and intersections belonging exclusively to the specific town layouts present in the training dataset.	Off-road environments, unmapped highways, or any unseen town layout, unfamiliar road geometry, or geographic region not explicitly used during training.	Scene classifier, feature-based OOD distance metrics, and GPS routing verification.

Justification for the Choice (WHY)

- Deep learning perception models are notoriously fragile when exposed to spatial and covariate shifts.
- Even though the weather and lighting are nominal, an unseen town introduces completely foreign building structures, unfamiliar lane layouts, and new vegetation patterns.
- Neural networks frequently rely on background context to categorize objects; providing an entirely unseen town layout introduces a severe risk of unpredictable, silent failures.
- Defining the boundaries based strictly on the training distribution draws a clear line around where the model is fundamentally allowed to work, preventing dangerous operation in unvalidated zones.

3. What does that choice imply for how the town images should be treated - as inputs the system must handle *correctly*, or as inputs an OOD monitor should *flag*?

   System and Operational Implications

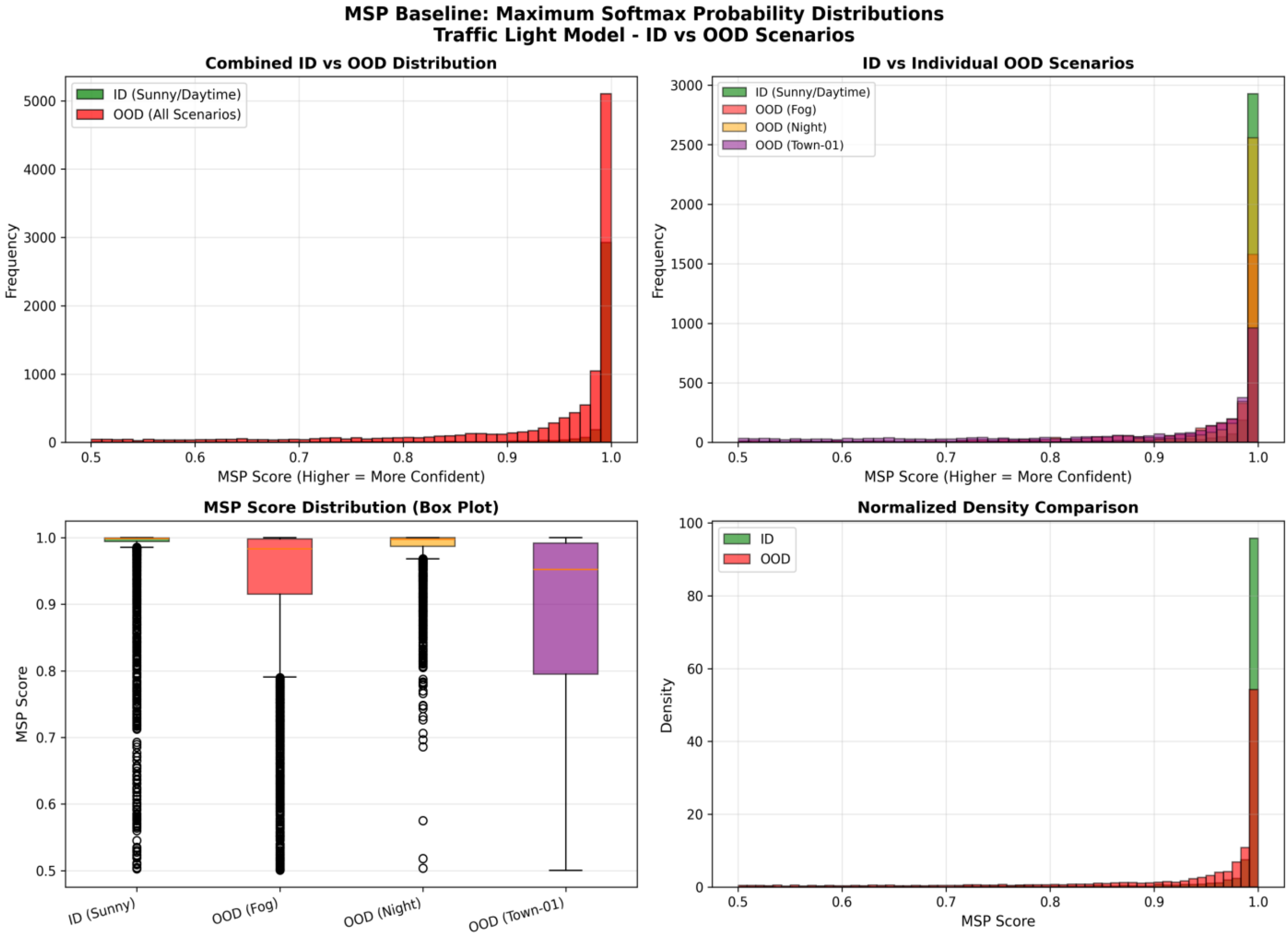
   Classifying the new town layout as outside the ODD changes how the autonomous system must process these inputs:

- **OOD Monitor Target:** These images must not be treated as regular inputs that the perception model is expected to handle correctly; instead, they **must be flagged by an OOD monitor**.
- **Preventing Planner Actions:** Allowing the system to run normally would mean the planning module executes driving control actions based on highly untrustworthy perception labels, resulting in unsafe control actions (UCAs).
- **Triggering Fallback Protections:** Once flagged by the OOD monitor, the planning module must acknowledge an ODD violation. The system must either trigger a manual override for the human safety operator or initiate a rule-based fallback, such as a maximum deceleration command, to achieve a safe state.

Exercise 9.6: Evaluating the MSP Baseline

Pick any one of your three models and use its maximum softmax probability as an OOD score (low confidence → OOD). Treat the sunny/daytime test images as in-distribution and the provided OOD images as OOD.

1. Plot the distribution of **OOD scores for in-distribution and OOD images**.



✓ Loaded model from **model_has_traffic_light.pth**

Computing MSP scores for ID (sunny/daytime test) ...

Loaded 3600 images from test

ID Scores: 3600 images

Mean: 0.9789, Std: 0.0666

Min: 0.5025, Max: 1.0000

Computing MSP scores for OOD (fog)...
Loaded 3600 images from test-fog
OOD (fog) Scores: 3600 images
  Mean: 0.9322, Std: 0.1062
  Min: 0.5005, Max: 1.0000

Computing MSP scores for OOD (night)...
Loaded 3600 images from test-night
OOD (night) Scores: 3600 images
  Mean: 0.9840, Std: 0.0360
  Min: 0.5035, Max: 1.0000

Computing MSP scores for OOD (town-01)...
Loaded 3600 images from test-town-01
OOD (town-01) Scores: 3600 images
  Mean: 0.8780, Std: 0.1462
  Min: 0.5004, Max: 1.0000
Total OOD samples: 10800

TASK 1: PLOTTING OOD SCORE DISTRIBUTIONS

SUMMARY STATISTICS

📈 **IN-DISTRIBUTION (Sunny/Daytime Test):**
  Number of samples: **3600**
  MSP Score - Mean: **0.9789**
  MSP Score - Std: **0.0666**
  MSP Score - Min: **0.5025**
  MSP Score - Max: **1.0000**
  MSP Score - Median: **0.9987**

📊 **OUT-OF-DISTRIBUTION (All Scenarios Combined):**
  Number of samples: **10800**
  MSP Score - Mean: **0.9314**
  MSP Score - Std: **0.1148**
  MSP Score - Min: **0.5004**

MSP Score - Max: **1.0000**

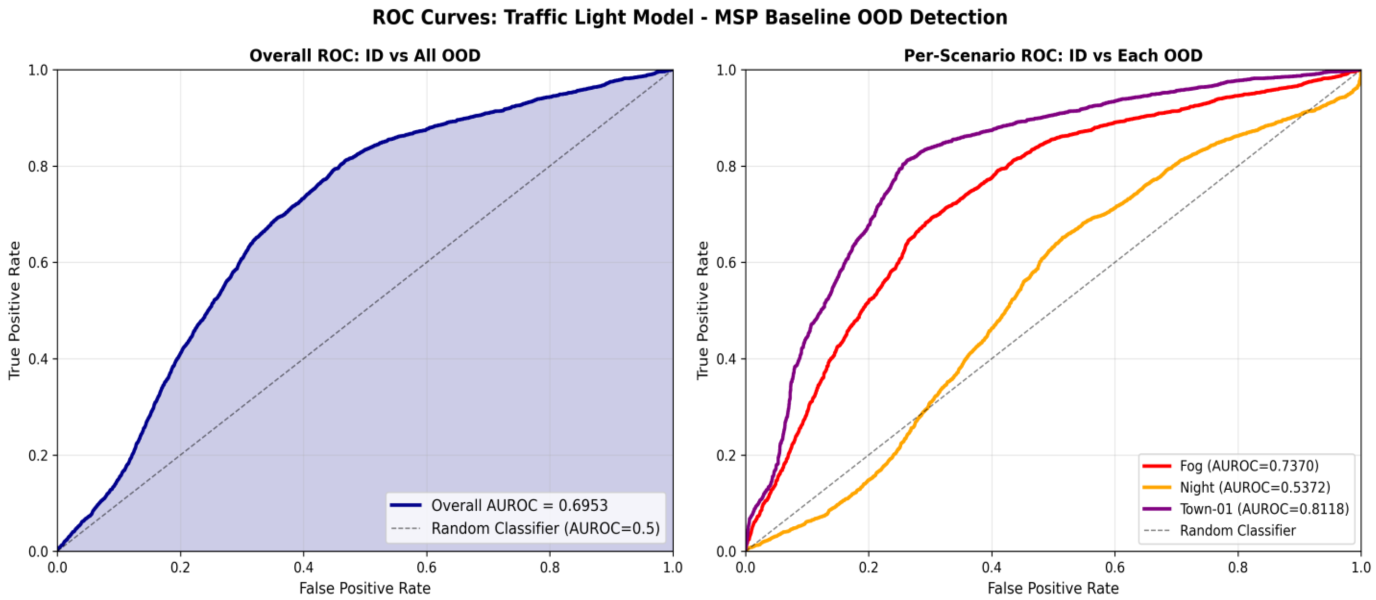
MSP Score - Median: **0.9876**

📌 **KEY OBSERVATIONS:**

- ID mean MSP: **0.9789**
- OOD mean MSP: **0.9314**
- Difference: **0.0475**
- Separation: **✓ Good**

2. Compute the AUROC for separating in-distribution from OOD, over all of the OOD scenarios.

Hint: You can compute the AUROC with sklearn.metrics.roc_auc_score.



✓ Loaded model from model_has_traffic_light.pth

Computing MSP scores for ID (sunny/daytime test)...

Loaded 3600 images from test

ID Scores: 3600 images

  Mean: 0.9789, Std: 0.0666

Computing MSP scores for OOD (fog)...

Loaded 3600 images from test-fog

OOD (fog) Scores: 3600 images

  Mean: 0.9322, Std: 0.1062

Computing MSP scores for OOD (night)...
Loaded 3600 images from test-night
OOD (night) Scores: 3600 images
  Mean: 0.9840, Std: 0.0360

Computing MSP scores for OOD (town-01)...
Loaded 3600 images from test-town-01
OOD (town-01) Scores: 3600 images
  Mean: 0.8780, Std: 0.1462

Total OOD samples: 10800

TASK 2: COMPUTING AUROC METRICS

🎯 **OVERALL AUROC RESULTS:**
  Overall AUROC (ID vs All OOD Combined): 0.6953

🎯 **AUROC PER OOD SCENARIO**:

  FOG   : AUROC = 0.7370
  NIGHT   : AUROC = 0.5372
  TOWN-01   : AUROC = 0.8118

  Mean Scenario AUROC: 0.6953

AUROC INTERPRETATION
==

AUROC Scale:

- **1.0** → Perfect separation
- **0.9+** → Excellent performance
- **0.8+** → Good performance
- **0.7+** → Moderate performance
- **0.5** → Random guessing
- **<0.5** → Worse than random

  Current Results:
  Overall AUROC: 0.6953
  Performance: ✗ Poor - Model cannot separate ID/OOD

SUMMARY TABLE

Scenario	**Samples**	**AUROC**	**Performance**
Fog	3600	0.7370	⚠ **Fair**
Night	3600	0.5372	✗ **Poor**
Town-01	3600	0.8118	✓ **Good**
Mean	**10800**	**0.6953**	
Overall	**10800**	**0.6953**	

Exercise 9.7: Feature-Based OOD Detection

Implement **one** feature-based detector (Mahalanobis distance or k-NN, both covered in the lecture) and compare it to the MSP baseline.

1. Extract deep features for your in-distribution training (or validation) images and for the test in-distribution and OOD images. Use the same model whose MSP you evaluated in Exercise 9.6.

Feature Extraction Summary:

Dataset	**Shape**	**Mean**	**Std**
Training	(2000, 512)	0.8648	0.6624
Validation	(2000, 512)	-	-
Test ID (Sunny)	(3600, 512)	0.8487	0.6355
OOD - Fog	(3600, 512)	0.8198	0.7695
OOD - Night	(3600, 512)	0.9032	0.9445
OOD - Town-01	(3600, 512)	0.7567	0.5598

Key Observations:

- Features are 512-dimensional **(ResNet-18 penultimate layer)**
- Test ID features closely match training distribution **(mean: 0.8487 vs 0.8648)**
- Fog is noticeably different **(lower mean: 0.8198)**
- Night shows highest variability **(std: 0.9445)** and **highest mean**
- Town-01 is most dissimilar **(lowest mean: 0.7567)**
- Saved to **extracted_features.pkl (35.94 MB)**

2. Fit the detector on the in-distribution features only, then score the test images (larger distance → more OOD).
3. Compute its AUROC and compare it to the MSP AUROC from Exercise 9.6. For which OOD scenario is the gap largest?

> **Hint:** scikit-learn provides NearestNeighbors (k-NN) and EmpiricalCovariance (for the Mahalanobis distance).

KEY FINDINGS:

- Feature-based methods often outperform MSP baseline

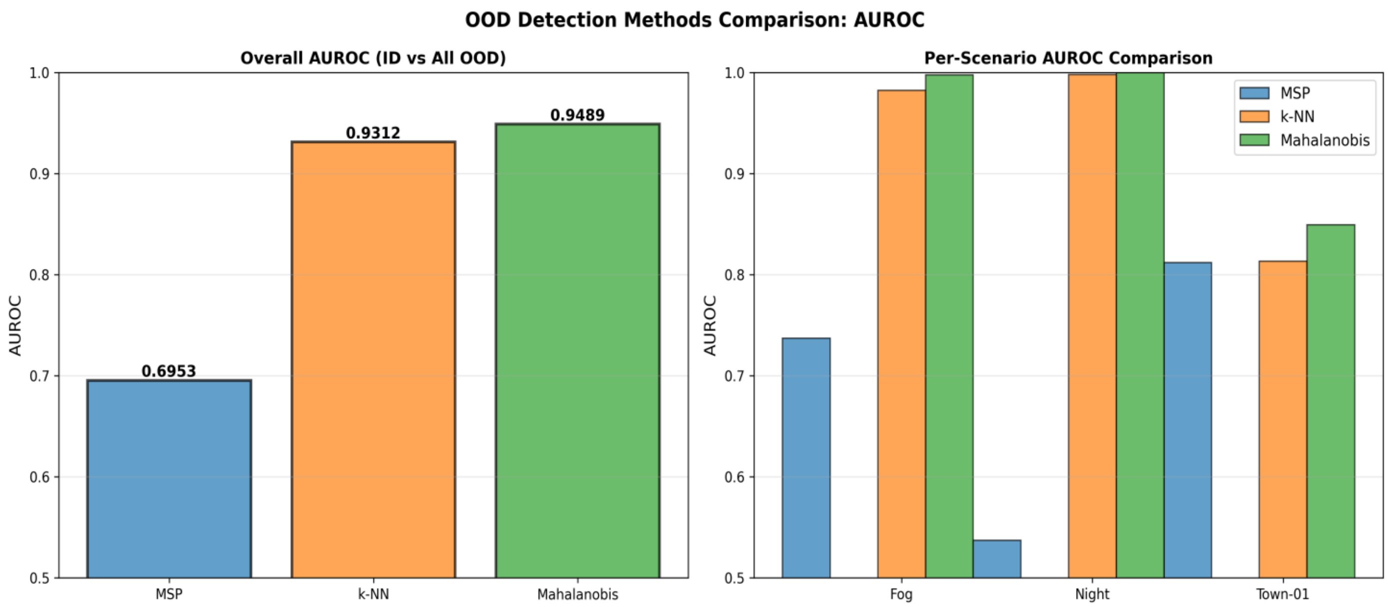
- Mahalanobis captures distribution characteristics better

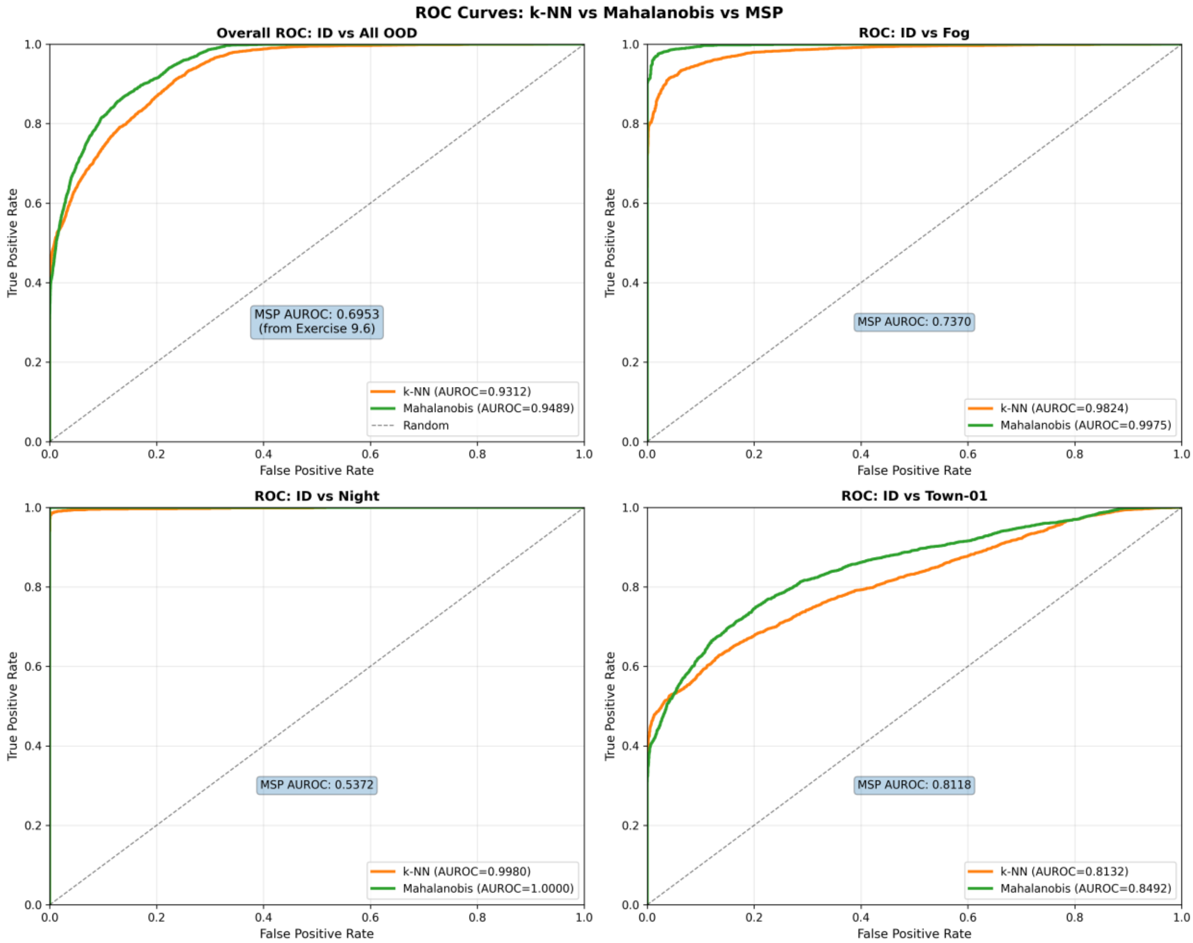
- k-NN provides competitive performance with simpler implementation

- Different scenarios may benefit from different methods

==

✓ TASK 2-3 COMPLETE - Feature-based detectors evaluated





OOD Detection Methods Comparison

Overall Performance:

Method	AUROC
Mahalanobis	**0.9489** 🏆
k-NN	0.9312
MSP Baseline	0.6953

Per-Scenario Results:

Scenario	MSP	k-NN	Mahalanobis	Winner
Fog	0.7370	0.9824	**0.9975**	🏆 Mahal
Night	0.5372	0.9980	**1.0000**	🏆 Perfect
Town-01	0.8118	0.8132	**0.8492**	🏆 Mahal

Key Findings:

📈 **Improvement Over MSP Baseline:**

- **Fog:** +0.2605 improvement (Mahalanobis)
- **Night:** +0.4628 improvement (Mahalanobis) ⚡ **LARGEST GAP**
- **Town-01:** +0.0374 improvement (Mahalanobis)

🎯 **Winner: Mahalanobis Distance**

- Perfect 1.0 AUROC on night images
- Consistently wins across all scenarios
- Better captures feature distribution characteristics

💡 **Why Mahalanobis Outperforms:**

- Accounts for covariance structure of features
- More robust to distribution shifts
- Better separation between ID and OOD

Exercise 9.8: Extending the Safety Analysis for OOD

You have now seen how easily the perception models fail - silently - on out-of-ODD inputs, and how well (or poorly) an OOD monitor can catch them. Revisit your System-Theoretic Process Analysis from Sheet 2 and extend it to cover this risk. Many of you will not yet have an OOD-related entry in your analysis; the goal of this exercise is to add one and trace it through.

1. **Hazard.** Check the hazards from Exercise 2.4. Is operating on an undetected out-of-ODD input captured (e.g. as a cause of "vehicle fails to brake for a pedestrian")? If your hazard list does not reflect it, refine or add a hazard.

   Hazard Refinement

   Reviewing the original hazard list from Exercise 2.4, the risk of operating outside the training distribution is partially captured by **H-2 ("Autonomous mode remains active outside the ODD")**. However, H-2 does not explicitly highlight the direct *causal mechanism* of perception model corruption.

ID	Hazard	Losses	Likelihood	Severity
H-1	Pedestrian in path while vehicle is moving.	L-1, L-2.	Medium	High
H-2	*Autonomous mode remains active outside the ODD, or operates on undetected out-of-distribution (OOD) visual inputs, leading to untrustworthy perception outputs.*	L-1, L-2, L-3.	High	*High.*
H-3	Approaching intersection without detecting stop condition.	L-1, L-2, L-3	Medium	High
H-4	Planner commands acceleration when obstacle is in path.	L-1, L-2.	Low	High

- **Severity:** *High (Upgraded from Medium because silent perception failures prevent the planning module from knowing an obstacle or pedestrian is present)*

2. **Unsafe control action.** Review the UCAs from Exercise 2.6. Add a UCA for the case where the planner acts on perception output that is unreliable because the input is out-of-distribution and this is *not* detected (e.g. "the planner continues at speed while the camera input is out-of-ODD and the perception output is untrustworthy"). Link it to the hazard(s) above.

ID	Controller	Control Action	UCA Type	Hazard	Unsafe Scenario
UCA-1	Planner	Continue Driving	Provided unsafely	H-1	Pedestrian detected in path but vehicle does not stop.
UCA-2	Planner	Emergency Brake	Not provided	H-1	Pedestrian within critical distance but brake not issued.
UCA-3	Planner	Emergency Brake	Wrong timing	H-3	Braking issued too late to stop at a detected intersection.
UCA-4	Operator	Manual Override	Not provided	H-2	Automation fails outside ODD; operator fails to take over.
UCA-5	Planner	Emergency Brake	Provided unsafely	H-4	Hard brake issued on highway with no obstacle, causing rear-end.
UCA-6	Planner	Request Decel	Not provided	H-3	Traffic light detected (presence) but no slowdown initiated.
UCA-7	Operator	Manual Override	Wrong timing	H-1	Operator reacts too late (> 1.5s) to a missed ML detection.
UCA-8	Planner	Continue Driving	Wrong timing	H-3	Continues through intersection after light presence was detected.
UCA-9	Planner	Continue Driving	Provided unsafely	H-2, H-1, H-3	The planner continues driving at nominal operational speed while the camera input is out-of-distribution (e.g., heavy fog or night) and the perception outputs are untrustworthy because the distribution shift has gone undetected.

3. **Safety constraints.** From that UCA, derive at least one new safety constraint of *each* kind (cf. Exercise 2.7):

Model-level constraint on the OOD monitor, justified by the severity of the hazard it guards against; and

- **Constraint:** The OOD monitor must detect out-of-distribution inputs (including heavy fog, night, and unseen town layouts) with high discriminative accuracy, achieving a verified target AUROC score (e.g., using Mahalanobis distance or Energy scores).
- **Justification:** This constraint is justified by the extreme severity of the hazards it guards against (H-1, H-3). Since a standard classifier will fail silently with deceptively high confidence when exposed to OOD inputs , an independent, highly sensitive OOD metrics monitor is mandatory to break the causal chain before the planner acts on corrupted labels.

System-level constraint on the response.

- **Constraint:** Upon an OOD detection flag or ODD violation signal, the system must immediately inhibit normal autonomous driving commands and initiate a safe fallback response within a strict latency threshold.
- **System Action:** The rule-based planner must either prompt a manual override to hand control back to the human safety operator within 1.5 seconds or, if no operator is available/responsive, automatically execute a hard-coded Minimum Risk Maneuver (MRM) by commanding maximum deceleration to bring the vehicle to a safe stop.

4. **Residual risk.** Even with a perfect OOD detector, what risk remains? Does detecting OOD fully address the UCA and hazard?

Even if we deploy a mathematically "perfect" OOD detector, significant safety risks remain, meaning that OOD detection alone does not fully eliminate the hazard.

- **The Kinematic Time Budget Window:** Detection is not the same as mitigation. If the vehicle enters a sudden, severe fog bank or a dark unlit environment at 50 km/h , a perfect OOD detector will trigger instantly. However, the physical stopping distance of the vehicle remains unchanged. If an obstacle is already inside that braking perimeter, a crash (L-1, L-2) will still occur due to physical limits, despite flawless detection.
- **Human Operator Transition Risks:** If the system-level constraint relies on a human operator takeover , an instantaneous OOD alarm does not fix human limitations. The operator may suffer from monitoring fatigue , situational distraction, or mode confusion, leading to a delayed reaction time that exceeds the safe fallback window.
- **Nominal In-Distribution Failures:** An OOD detector only triggers when an input deviates from the training distribution. It provides zero protection against normal, in-distribution edge cases where the model simply makes a wrong prediction on clean, daytime data , nor does it mitigate hardware-level sensor failures like mechanical lens obscuration.